Literals In java

In Java, a literal is a fixed, constant value that is represented directly in the source code. Literals are used to assign values to variables and represent data that does not change during program execution.

Java supports several types of literals, which correspond to different data types:

- Integer Literals: Represent whole numbers without a fractional part. By default, they are of type `int`.
  - Decimal (base 10): Standard numbers (e.g., `100`, `42`).
  - Octal (base 8): Must be prefixed with a `0` (e.g., `075`, which is 61 in decimal).
  - Hexadecimal (base 16): Must be prefixed with `0x` or `0X` and can include digits 0-9 and letters A-F (e.g., `0x1A`, which is 26 in decimal).
  - Binary (base 2): Introduced in Java 7, must be prefixed with `0b` or `0B` (e.g., `0b1010`, which is 10 in decimal).
  - To specify a `long` literal, append the suffix `L` or `l` (e.g., `7500000000L`).
- Floating-Point Literals: Represent numbers with fractional parts (real numbers). By default, they are of type `double`.
  - Standard notation: (e.g., `3.14`, `-2.5`).
  - Scientific notation: Uses `e` or `E` to indicate an exponent (e.g., `1.2e3` represents $1.2 * 10^3$).
  - To specify a `float` literal, append the suffix `F` or `f` (e.g., `3.14f`). The suffix `D` or `d` can be used for `double`, but it is optional.
- Character Literals: Represent a single Unicode character and are enclosed in single quotes (e.g., `'A'`, `'1'`, `'$'`). They can also use:
  - Escape sequences: Special combinations for non-graphic characters like newline (`\n`), tab (`\t`), backslash (`\\`), or single quote (`\'`).
  - Unicode representation: Using the format `\uXXXX` where `XXXX` is a four-digit hexadecimal value (e.g., `\u0041` for 'A').

- String Literals: Represent a sequence of characters and are enclosed in double quotes (e.g., `"Hello, Java!"`, `"123"`). String literals are objects of the `String` class and can also use escape sequences. Java 13+ introduced text blocks for multiline strings using triple double-quotes (""").
- Boolean Literals: Represent logical values and have only two possible values: `true` and `false`. These are case-sensitive and must be in lowercase.
- Null Literal: A special literal with the single value `null`, which indicates that a reference variable does not point to any object. It can be assigned to any reference type (like `String` or custom classes), but not to primitive types.